

Customer Churn Analysis – Telecom Industry

1. Introduction

In today's highly competitive telecom industry, customer retention is more crucial than ever. Understanding the factors that lead to customer churn can help telecom companies retain valuable customers and improve profitability. This project leverages machine learning techniques to analyze customer behavior and predict churn, enabling data-driven decisions for customer engagement and retention strategies.

2. Abstract

This project involves the end-to-end analysis of customer churn using a publicly available telecom dataset. The analysis includes preprocessing, model building using a Random Forest classifier, and using SHAP (SHapley Additive exPlanations) for explainability. Key churn predictors were identified and customers were segmented into Loyal, At Risk, and Dormant categories. The insights drawn offer actionable recommendations for improving customer retention and reducing churn.

3. Tools Used

- **Python Libraries:** Pandas, NumPy, Scikit-learn, SHAP
 - **Jupyter Notebook:** Development Environment
 - **SQL:** For data querying and analysis
 - **PowerPoint:** For presentation and visual reporting
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4. Steps Involved in Building the Project

1. Data Preprocessing

- Checked for null values and handled missing data
- Encoded categorical variables using Label Encoding
- Scaled numerical features for model readiness

2. Train-Test Split & Model Training

- Dataset was split (80-20 ratio)
- Trained a Random Forest Classifier

3. Model Evaluation

- Evaluated using Accuracy, Precision, Recall, F1-Score
- Achieved good balance of precision and recall for churn prediction

4. Explainability using SHAP

- Applied SHAP to interpret feature contributions
- Identified key drivers of churn like contract type, tenure, and internet service

5. Customer Segmentation

- Segmented users based on churn probability:
 - **Loyal Customers** (Low probability)
 - **At Risk Customers** (Moderate probability)
 - **Dormant/High Risk Customers** (High probability)
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5. Conclusion

The model successfully predicts customer churn and provides valuable insights into churn drivers using SHAP. By identifying customers at risk, telecom companies can take proactive measures such as personalized offers, rewards for loyalty, or service improvements. This project exemplifies how machine learning can be utilized effectively in business for customer retention strategies.
